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Data Mining
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Recent Graduates

Dataset Narrative

Tentatively I may call the project Recent Graduates. It was suggested that we pick something from our major, but my major is Data Science so I just tried to pick a subject that was somewhat interesting to me. This dataset contains information about recent graduates, including their major, their employment, their gender, and their earnings. Each observation in the dataset is of a different major. This dataset was last updated 7 years ago. According to the “readme” markdown document for the data on data.world, the data was apparently used in a piece by FiveThirtyEight on “the earnings of college majors.” The csv I pulled from is for a table with 21 attributes so I plan to eventually create a subset with just 15 of the attributes. For now I’ll give variable descriptions for all 21 attributes in the Data Sources section. From this dataset I expect to find that graduates who have jobs that require college degrees are financially better off than graduates who have jobs that do not require a college degree.

Audience

If I wanted to give a presentation on this data, the expected audience would probably be recent graduates or people graduating soon, as I think they’d be the ones most interested in what to expect financially post-graduation. It’s also possible that people who have not gone to college could use this data to help inform their decision on whether or not to pursue a college degree, as they can compare, among other things, the number of people who have jobs requiring a college degree to the number of people with jobs that don’t have a college degree and the number of people in a given major who have found employment. This data is interesting to me personally because I’ve been job searching as a graduate for some time now with limited success.

Data Sources

I found this dataset on data.world after clicking on their “education” data topic; from the list of datasets I picked one called “College Majors” and from a group of csvs I picked out the one called “recent-grads.csv.” (Full disclosure Dr. Rhodes, I did discover that a copy of the data exists on Kaggle. It technically originates from the Census Bureau though. If you do have a

problem with it being accessible on Kaggle just let me know and I can try to pick something else.)

I used the `glimpse` function in RStudio to create snippets of the table. Because of large value lengths some attributes like Major display less than five values, but I provided a snippet of the head of the data, a snippet of the tail of the data and a random sample of the data to help give a sense of what the dataset looks like, so I figured it wasn't too important.

```
recent_grads%>%  
  head()%>%  
  glimpse()  
  
## Rows: 6  
## Columns: 21  
## $ Rank                <dbl> 1, 2, 3, 4, 5, 6  
## $ Major_code          <dbl> 2419, 2416, 2415, 2417, 2405, 2418  
## $ Major               <chr> "PETROLEUM ENGINEERING", "MINING AND  
MINERAL ENGI...  
## $ Major_category      <chr> "Engineering", "Engineering",  
"Engineering", "Eng...  
## $ Total               <dbl> 2339, 756, 856, 1258, 32260, 2573  
## $ Sample_size         <dbl> 36, 7, 3, 16, 289, 17  
## $ Men                 <dbl> 2057, 679, 725, 1123, 21239, 2200  
## $ Women               <dbl> 282, 77, 131, 135, 11021, 373  
## $ ShareWomen          <dbl> 0.1205643, 0.1018519, 0.1530374,  
0.1073132, 0.341...  
## $ Employed            <dbl> 1976, 640, 648, 758, 25694, 1857  
## $ Full_time           <dbl> 1849, 556, 558, 1069, 23170, 2038  
## $ Part_time           <dbl> 270, 170, 133, 150, 5180, 264  
## $ Full_time_year_round <dbl> 1207, 388, 340, 692, 16697, 1449  
## $ Unemployed          <dbl> 37, 85, 16, 40, 1672, 400  
## $ Unemployment_rate   <dbl> 0.01838053, 0.11724138, 0.02409639,  
0.05012531, 0...  
## $ Median              <dbl> 110000, 75000, 73000, 70000, 65000,  
65000  
## $ P25th               <dbl> 95000, 55000, 50000, 43000, 50000,  
50000  
## $ P75th               <dbl> 125000, 90000, 105000, 80000, 75000,  
102000  
## $ College_jobs        <dbl> 1534, 350, 456, 529, 18314, 1142  
## $ Non_college_jobs    <dbl> 364, 257, 176, 102, 4440, 657  
## $ Low_wage_jobs       <dbl> 193, 50, 0, 0, 972, 244
```

```

recent_grads%>%
  tail()%>%
  glimpse()

## Rows: 6
## Columns: 21
## $ Rank <dbl> 168, 169, 170, 171, 172, 173
## $ Major_code <dbl> 3302, 3609, 5201, 5202, 5203, 3501
## $ Major <chr> "COMPOSITION AND RHETORIC", "ZOOLOGY",
"EDUCATION..."
## $ Major_category <chr> "Humanities & Liberal Arts", "Biology
& Life Scie..."
## $ Total <dbl> 18953, 8409, 2854, 2838, 4626, 1098
## $ Sample_size <dbl> 151, 47, 7, 13, 21, 2
## $ Men <dbl> 14440, 3050, 522, 568, 931, 134
## $ Women <dbl> 28809, 5359, 2332, 2270, 3695, 964
## $ ShareWomen <dbl> 0.6661194, 0.6372934, 0.8170988,
0.7998591, 0.798...
## $ Employed <dbl> 15053, 6259, 2125, 2101, 3777, 742
## $ Full_time <dbl> 10121, 5043, 1848, 1724, 3154, 593
## $ Part_time <dbl> 6612, 2190, 572, 648, 965, 237
## $ Full_time_year_round <dbl> 7832, 3602, 1211, 1293, 2738, 410
## $ Unemployed <dbl> 1340, 304, 148, 368, 214, 87
## $ Unemployment_rate <dbl> 0.08174221, 0.04632028, 0.06511219,
0.14904820, 0...
## $ Median <dbl> 27000, 26000, 25000, 25000, 23400,
22000
## $ P25th <dbl> 20000, 20000, 24000, 25000, 19200,
20000
## $ P75th <dbl> 35000, 39000, 34000, 40000, 26000,
22000
## $ College_jobs <dbl> 4855, 2771, 1488, 986, 2403, 288
## $ Non_college_jobs <dbl> 8100, 2947, 615, 870, 1245, 338
## $ Low_wage_jobs <dbl> 3466, 743, 82, 622, 308, 192

recent_grads%>%
  sample_n(6, TRUE)%>%
  glimpse()

## Rows: 6
## Columns: 21
## $ Rank <dbl> 26, 86, 144, 61, 40, 60

```

```
## $ Major_code      <dbl> 2406, 5004, 1105, 6108, 5102, 6210
## $ Major           <chr> "CIVIL ENGINEERING", "GEOLOGY AND
EARTH SCIENCE",...
## $ Major_category  <chr> "Engineering", "Physical Sciences",
"Agriculture ..."
## $ Total           <dbl> 53153, 10972, 7416, 23551, 2116, 25894
## $ Sample_size     <dbl> 565, 78, 110, 38, 31, 260
## $ Men             <dbl> 41081, 5813, 5079, 25463, 7575, 1749
## $ Women           <dbl> 12072, 5159, 7841, 20957, 5727, 690
## $ ShareWomen      <dbl> 0.2271179, 0.4701969, 0.6068885,
0.4514649, 0.430...
## $ Employed        <dbl> 43041, 8296, 6594, 16620, 1778, 19660
## $ Full_time       <dbl> 38302, 6966, 5798, 12537, 1392, 17563
## $ Part_time       <dbl> 10080, 2913, 1246, 5346, 579, 4890
## $ Full_time_year_round <dbl> 29196, 5008, 4522, 9131, 1115, 12823
## $ Unemployed      <dbl> 3270, 677, 314, 977, 137, 2092
## $ Unemployment_rate <dbl> 0.07060957, 0.07544857, 0.04545454,
0.05552083, 0...
## $ Median          <dbl> 50000, 36200, 32000, 40000, 46000,
40000
## $ P25th           <dbl> 40000, 28000, 22900, 20000, 38000,
30000
## $ P75th           <dbl> 60000, 47000, 40000, 90000, 53000,
50000
## $ College_jobs    <dbl> 28526, 4858, 2089, 11573, 162, 3383
## $ Non_college_jobs <dbl> 9356, 2792, 3545, 4493, 1475, 9482
## $ Low_wage_jobs   <dbl> 2899, 959, 1231, 1121, 124, 3046
```

Name	Type	Range	Description
Rank	Ordinal, discrete.		An assigned ranking based on the median earnings of graduates who work full-time and year-round.
Major_code	Nominal, categorical.		An identifier for the major.

Name	Type	Range	Description
Major	Nominal, categorical		The name of the major, e.g. Petroleum Engineering.
Major_category	Nominal, categorical		Gives a broader field the major falls into, e.g. “Engineering” instead of “Petroleum Engineering.”
Total	Ratio, discrete	Non-negative.	The number of graduates with the major.
Sample_size	Ratio, discrete	Non-negative.	The number of graduates who work full-time and year-round used to compute Median, P25th and P75th.
Men	Ratio, discrete	Non-negative.	The number of men who graduated with the major.
Women	Ratio, discrete	Non-negative.	The number of women who graduated with the major.
ShareWomen	Ratio, continuous	0 to 1.	The decimal proportion of graduates in the major who are women.
Employed	Ratio, discrete	Non-negative.	The number of graduates in the major who are employed, based on “ESR.”
Full_time	Ratio, discrete	Non-negative.	The number of graduates in the major who work 35 hours a week or more.

Name	Type	Range	Description
Part_time	Ratio, discrete	Non-negative.	The number of graduates in the major who work less than 35 hours a week (unemployed graduates do not count towards the total).
Full_time_year_round	Ratio, discrete	Non-negative.	The number of graduates in the major who work at least 35 hours a week and at least 50 weeks a year based on “WKW” and “WKHP.”
Unemployed	Ratio, discrete	Non-negative.	The number of graduates in the major who are unemployed, based on “ESR.”
Unemployment_rate	Ratio, continuous	0 to 1.	The decimal proportion of graduates in the major who are unemployed; derived from Employed and Unemployed based on the formula $\text{Unemployed} / (\text{Unemployed} + \text{Employed})$.
Median	Ratio, discrete	Non-negative.	The median annual income of graduates in the major who work full-time and year-round.

Name	Type	Range	Description
P25th	Ratio	Non-negative.	Same as above but it's the 25th percentile not the 50th percentile.
P75th	Ratio	Non-negative.	Same again but it's the 75th percentile.
College_jobs	Ratio	Non-negative.	The number of graduates in the major who have a job that requires a college degree.
Non_college_jobs	Ratio	Non-negative.	The number of graduates in the major who have a job that does not require a college degree.
Low_wage_jobs	Ratio	Non-negative.	The number of graduates in the major who have a "low-wage service job."

URLs

Note: when you use the Census Bureau link for the page where the data was originally pulled a 404 message pops up. But below that message you should see a hyperlink for the American Community Survey (ACS) page which I believe is the survey that was used to collect the data. I also included a link for the College Majors dataset on data.world, and a github link for FiveThirtyEight that was provided on that data.world page.

<https://data.world/fivethirtyeight/college-majors>

<https://github.com/fivethirtyeight/data>

http://www.census.gov/acs/www/data_documentation/pums_data/